CampusAI

Object Oriented Analysis and Design (OOAD)

Sample Study Notes • For learning and revision

1. Introduction

- Object Oriented Analysis and Design (OOAD) is a software development approach that models a system using objects and classes.
- Analysis focuses on understanding what the system must do. Design focuses on how the system will be built.

2. Important Concepts

- Object: An entity with state (data) and behaviour (methods).
- Class: A blueprint used to create objects.
- Encapsulation: Bundling data and methods and controlling access to data.
- Abstraction: Showing essential features while hiding unnecessary details.
- Inheritance: Creating a new class from an existing class.
- Polymorphism: Allowing the same operation to behave differently in different situations.

3. UML Diagrams

- Use-case diagram: Shows actors and the functions they use.
- Class diagram: Shows classes, attributes, methods and relationships.
- Sequence diagram: Shows messages exchanged over time.
- Activity diagram: Represents the workflow of a process.

4. Example

- In a college system, Student can be a class. A student object may contain a name and roll number and methods such as viewMarks() and registerCourse().

5. Quick Revision Questions

- Differentiate a class and an object.
- Explain encapsulation and inheritance.
- What is the purpose of a use-case diagram?
- What information is shown in a sequence diagram?

Note: These are introductory sample notes. Check your course syllabus and class materials for the exact topics and terminology.